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Optimizing Power, Performance, and Signal Integrity in Sub-6nm Physical Design Architectures

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***Abstract*— As semiconductor manufacturing advances into sub-6 nm territories, IR-drop analysis has become increasingly critical for ensuring the reliability and performance of integrated circuits. This paper outlines a holistic physical-design methodology for a memory subsystem, aimed at mitigating power consumption, thermally induced effects, and signal-integrity issues in ultra-scaled, high-speed VLSI systems. Leveraging mixed-threshold voltage (VT) optimization, H-tree clock distribution networks, and advanced placement strategies, this approach effectively mitigates IR drop, enhances signal integrity, and optimizes the power-performance-area (PPA) trade-off. The result is a high-speed, energy-efficient, and resilient VLSI design tailored to meet the stringent demands of sub-6nm technology nodes.**

Keywords—VLSI, Power Optimization, IR Drop, Signal Integrity, Physical Design, PPA Optimization, Static Power, Dynamic Power.

I. INTRODUCTION

In semiconductor design, the physical layout is pivotal in optimizing power efficiency, performance, and form factor. The triad of Power, Performance, and Area (PPA) serves as the primary metric for evaluating VLSI physical design success. The challenge lies in harmonizing these often-conflicting parameters to produce cost-effective, high-quality integrated circuits. Power optimization focuses on reducing energy consumption, a critical factor in battery-powered devices and energy-efficient computing. Performance optimization aims to meet or exceed speed and timing requirements, essential for applications demanding high computational throughput. Area optimization seeks to minimize chip size, directly impacting manufacturing costs and device portability. The interplay between these factors creates a complex optimization landscape. For instance, increasing a chip's clock speed often necessitates higher power consumption and potentially larger area.

Reducing chip size can lead to trade-offs, such as diminished performance or increased power density. Achieving an optimal balance among these interdependent factors—power, performance, and area (PPA)—is essential for developing cost-effective and high-performance integrated circuits. The escalating demand for integrated circuits that seamlessly integrate low power consumption, high performance, and compact form factors spans a wide array of applications, including wearable devices, Internet of Things (IoT) sensors, data centers, and 5G infrastructure. In response to these challenges, the semiconductor industry has pioneered a suite of innovative design techniques, methodologies, and tools specifically tailored for PPA optimization in VLSI physical design. These advancements encompass:

Multi-objective optimization is essential in VLSI design to achieve efficient and balanced solutions that meet the stringent requirements of power, performance, and area.

Leveraging machine learning methodologies to forecast and enhance power, performance, and area (PPA) metrics during the initial stages of the design cycle.

Incorporating advanced power gating and clock gating strategies enables fine-grained power management in VLSI designs, effectively reducing both dynamic and static power consumption.

Incorporating three-dimensional integrated circuit (3D IC) architecture and through-silicon vias (TSVs) facilitate enhanced performance and a reduced footprint in advanced VLSI designs. By vertically stacking multiple semiconductors dies and interconnecting them using TSVs, these technologies enable shorter interconnects, which lead to faster signal propagation and lower power consumption.

Implementing adaptive voltage scaling (AVS) and dynamic frequency scaling (DFS) techniques enable real-time optimization of power, performance, and area (PPA) metrics in VLSI designs. AVS dynamically adjusts the supply voltage to match the chip's performance requirements under varying workloads, compensating for process variations, temperature fluctuations, and aging effects. This closed-loop approach ensures that each processor domain operates at the minimum voltage necessary, leading to significant power savings.

As semiconductor technology advances beyond the 5nm node and new computing paradigms like quantum and neuromorphic systems emerge, optimizing Power, Performance, and Area (PPA) becomes increasingly critical. PPA optimization is essential for developing compact, high-performance integrated circuits tailored to meet the diverse needs of modern applications.

This paper provides a comprehensive exploration of PPA optimization in VLSI physical design. Section II reviews the current state-of-the-art techniques in PPA optimization. Section III outlines the objectives of this study. Section IV presents our proposed methodology for achieving optimal PPA balance. Section V discusses the results obtained from our approach. Finally, Section VI concludes the paper and suggests directions for future research in this rapidly evolving field.

II. LITERATURE REVIEW

Recent years have seen significant progress in optimizing Power, Performance, and Area (PPA) within VLSI physical

design. This section highlights the latest research trends and advancements in this area.

A. Machine Learning Approaches

Machine learning has emerged as an effective approach for enhancing PPA optimization. In [1], a machine learning-based framework was presented for power-aware floor planning, resulting in significant gains in power efficiency. Similarly, [2] developed a machine learning-driven placement framework tailored for mixed-size designs, underscoring the increasing influence of AI in the field of physical design optimization.

Machine learning-based technique for low-power clock tree synthesis, addressing the significant challenge of minimizing power consumption in clock distribution networks [3]. Expanding this approach, [4] utilized machine learning for post-layout dynamic power optimization, contributing to enhanced efficiency in VLSI circuit design.

B. Multi-Objective Optimization

Recognizing the interdependence of power, performance, and area, researchers have developed multi-objective optimization strategies. [5] proposed a power-aware routing method that addresses multiple design objectives simultaneously, achieving a balanced trade-off between power consumption and other key metrics in VLSI circuits.

C. 3D IC Design

The rise of 3D IC technology has introduced unique challenges and opportunities for PPA optimization. In response, [6] proposed a thermal-aware 3D placement strategy focused on enhancing power efficiency while managing the specific thermal issues inherent to 3D ICs. Likewise, [7] developed an integrated power optimization framework for 3D ICs, addressing both thermal concerns and power delivery limitations.

D. Post-Layout Optimization

Post-layout optimization techniques have gained prominence for their ability to accurately enhance PPA metrics. For instance, [8] introduced a method for post-layout leakage power optimization, targeting a key source of power consumption in modern VLSI circuits.

E. Emerging Techniques

Several emerging techniques show considerable promise for PPA optimization. For instance, [9] investigated reinforcement learning for power-aware routing, demonstrating the impact of advanced AI methods in physical design. Likewise, [10] proposed a power-aware placement technique that utilizes graph convolutional networks, capitalizing on the advantages of graph-based deep learning models.

In summary, the field of PPA optimization in VLSI physical design is advancing quickly, with machine learning, multi-objective optimization, and 3D IC design emerging as major areas of innovation. These developments play a vital role in addressing the growing need for energy-efficient and high-performance integrated circuits across a wide range of applications.

III. Problem Statement and Objectives

The main challenge is to address EM/IR and other thermally induced effects at smaller technology nodes, while simultaneously achieving power efficiency in high-speed integrated circuits. The specific objectives are as follows:

1) Identify and optimize current logic area bottlenecks for block level sub-system.
2) Improve timing margins for critical interfaces in Signal Integrity mode for Block-Level SoC Design
3) Reduce power in SoC Design by applying mix-VT optimization and varying different metal layers during routing.
4) Achieve up to 20% reduction in IR drop in the subsystem with 20M gate count
5) Resolve Signal Integrity issues (PGEM, SEM) while meeting timing and area requirements
6) Perform Physical Verification checks on the subsystem

IV. Methodology

A comprehensive methodology is employed, encompassing various stages of physical design and the techniques used to meet PPA and Signal Integrity requirements.

A. Sanity Checks

To verify the accuracy of inputs from the library and synthesis netlist, these checks are essential. Skipping them can lead to issues in later design stages. Essentially, this process involves validating design consistency by performing SDC (System Design Constraint) checks and library verifications to ensure the files are complete and free of errors.

B. Floor Planning

Floor planning is a key stage in chip design that involves strategically allocating space and roughly positioning major functional blocks within the integrated circuit. This phase is essential for meeting performance and area targets and lays the foundation for the following placement and routing steps. Macros are substantial, self- contained units that function as black boxes. While their logical and electronic behaviors are defined, the internal structure may or may not be fully disclosed. The challenge of macro-cell placement lies in optimally positioning these components to achieve specific objectives, such as minimizing total wire length or optimizing timing. Several important principles should be considered when approaching macro placement:

- Functional Grouping: Organize macros based on the design hierarchy to enhance efficiency.
- I/O Port Consideration: Consider the locations of input and output ports when positioning macros.
- Edge Placement: Prioritize placing macros along the core boundary to reduce routing obstacles and minimize detours.
- Inter-Macro Spacing: Allow sufficient space between macro-cells depending on the available routing resources to accommodate necessary buffering.
- Pin Connectivity and Orientation: Carefully consider the connections between pins and the orientation of macros to optimize overall layout.

Adhering to these guidelines it improves performance and streamlines implementation in subsequent design phases.

C. Placement

The placement stage plays a vital role in maximizing chip performance and efficient use of area. Placement decisions are guided by various factors such as timing requirements, congestion management, and power optimization. This process involves strategically positioning individual standard cells within the core area to meet timing, power constraints and minimize overall area. One key strategy for improving placement quality is the utilization of placement blockages and cell padding, particularly for clock cells. Placement blockages when applied allow only a certain percentage of the area to be used to place the standard cells thereby helping in reducing congestion and local hotspots.

Cell padding around clock cells is an effective technique for minimizing IR drop. Because clock networks switch often, they contribute substantially to dynamic power consumption in chips. Implementing padding around clock buffers, inverters, and clock gate cells prevents the close placement of standard or densely packed logic cells, thereby helping to alleviate power-related challenges.

By combining placement bounds for core logic and cell padding for clock cells, we can achieve more robust designs with reduced IR drop concerns. This methodology not only addresses power integrity issues but also signal integrity problems.

D. Clock tree synthesis

The clock tree synthesis (CTS) plays a vital role in improving chip performance and affecting timing , particularly in advanced process nodes with many signoff corners. The steps of design for floor planning , power planning and placement take into account the CTS to avoid timing violations. Under the multi-mode-multi-corner (MMMC) design, the clock signal has many diversities , resulting in excessive clock skew, which is finally reflected in many timing violations and limits the performance of the chip.

The Cadence flexible H-tree feature provides the electrically symmetric buffering and balanced wire length benefits of an H-Tree to reduce timing violation paths and achieve low skew, and relaxes the requirement to be geometrically symmetric, enabling automated synthesis even in floorplans with placement restrictions. D8 buffers are used for leaf nodes and D16 buffers for trunks.

Routing Considerations for CTS The clock frequency of the block is up to 1.6 GHz. Considering the influence of electromigration and crosstalk on the timing path, it is necessary to define the non-default routing rules for CTS. The recommendation is to use the highest possible routing layers and wider than default width in lower layers. Because the highest routing layer has low resistance , high electrical mobility and strong connectivity.

E. Signal Integrity analysis

Dynamic IR drop analysis plays a vital role in the design and validation of modern integrated circuits. The two primary methodologies for conducting such analysis: vector-less and vector-based approaches. Vector-less Dynamic IR Analysis: This approach utilizes average chip power or toggling rates derived from average power estimations. While computationally efficient, it suffers from potential inaccuracies due to the variability of average power across different operational scenarios. The method often requires multiple iterations during final design stages, potentially impacting time- to-market. Vector-based Dynamic Analysis: This more sophisticated approach employs gate-level Value Change Dump (VCD) files for enhanced accuracy. It can be implemented using :

- True-timing gate-level modeling (gate to VCD)
- Zero delay Gate-VCD (GVCD)
- RTL-Verilog simulation analysis (RTL-VCD)

The gate-VCD method, derived from synthesis netlists, is preferred for dynamic IR drop sign-off due to its higher fidelity. Analysis Coverage: Optimal analysis coverage is achieved at approximately 50%, representing the switching of half the instances. The coverage metric is defined as: Coverage = (Number of switching instances) / (Total number of instances) Higher accuracy in IR drop analysis correlates with coverage approaching 50%.

While vector-less analysis offers simplicity and computational efficiency, vector-based analysis using gate-level VCD files provides superior accuracy for dynamic IR drop analysis. This is particularly crucial when considering realistic switching scenarios and aiming for com- prehensive coverage in advanced IC designs.

V. Results and Discussion

A. IR drop Reduction

Table I shows the value of IR drop before applying the optimization techniques. The IR drop seen was 16mV. Table II shows the value of IR drop after applying partial placement blockages and cell padding during the placement stage. The IR drop value reduced to 13mV. A 19% reduction in IR drop was achieved through the strategic use of partial placement blockages to alleviate local congestion hotspots.

TABLE I. IR Drop before placement optimization

loc x	loc y	ideal V	min pg	min vdd	max vss	pg arc
540.38	269.56	0.725	0.709	0.717	0.008	VDD/VSS
539.04	269.56	0.725	0.710	0.717	0.007	VDD/VSS
538.35	270.57	0.725	0.711	0.717	0.007	VDD/VSS
539.33	269.22	0.725	0.710	0.717	0.007	VDD/VSS
540.29	270.57	0.725	0.710	0.717	0.007	VDD/VSS
543.65	191.48	0.725	0.710	0.717	0.007	VDD/VSS
539.81	269.56	0.725	0.710	0.717	0.007	VDD/VSS
540.67	270.23	0.725	0.711	0.718	0.007	VDD/VSS
541.54	269.56	0.725	0.711	0.718	0.007	VDD/VSS
537.22	465.59	0.725	0.711	0.718	0.007	VDD/VSS
539.18	270.57	0.725	0.711	0.719	0.008	VDD/VSS
540.67	269.56	0.725	0.711	0.719	0.008	VDD/VSS
538.85	270.23	0.725	0.711	0.718	0.007	VDD/VSS
540.53	269.56	0.725	0.712	0.719	0.007	VDD/VSS
539.42	269.89	0.725	0.712	0.718	0.006	VDD/VSS
540.24	268.88	0.725	0.712	0.718	0.007	VDD/VSS
543.31	191.48	0.725	0.712	0.719	0.007	VDD/VSS
600.24	318.23	0.725	0.712	0.719	0.007	VDD/VSS
543.31	191.81	0.725	0.712	0.718	0.006	VDD/VSS
540.14	165.45	0.725	0.712	0.719	0.007	VDD/VSS
544.41	166.19	0.725	0.712	0.719	0.007	VDD/VSS
545.95	166.13	0.725	0.712	0.719	0.007	VDD/VSS
687.26	368.93	0.725	0.712	0.718	0.006	VDD/VSS
543.31	192.15	0.725	0.712	0.718	0.006	VDD/VSS
543.22	191.48	0.725	0.712	0.719	0.007	VDD/VSS

TABLE III. IR DROP AFTER PLACEMENT OPTIMIZATION

loc x	loc y	ideal V	min pg	min vdd	max vss	pg arc
257.33	475.26	0.725	0.712	0.718	0.007	VDD/VSS
612.24	312.23	0.725	0.713	0.719	0.006	VDD/VSS
532.59	211.41	0.725	0.713	0.719	0.006	VDD/VSS
641.34	183.39	0.725	0.713	0.719	0.006	VDD/VSS
236.91	363.91	0.725	0.713	0.719	0.006	VDD/VSS
424.53	437.57	0.725	0.713	0.719	0.006	VDD/VSS
576.37	341.48	0.725	0.713	0.720	0.006	VDD/VSS
794.39	323.57	0.725	0.714	0.719	0.006	VDD/VSS
575.79	398.48	0.725	0.714	0.720	0.006	VDD/VSS
329.76	387.23	0.725	0.714	0.719	0.006	VDD/VSS
642.90	342.59	0.725	0.714	0.719	0.006	VDD/VSS
242.24	378.98	0.725	0.714	0.719	0.006	VDD/VSS
942.50	211.07	0.725	0.714	0.719	0.006	VDD/VSS
742.59	213.74	0.725	0.714	0.719	0.006	VDD/VSS
593.07	298.74	0.725	0.714	0.719	0.005	VDD/VSS
359.14	147.14	0.725	0.714	0.719	0.006	VDD/VSS
385.65	325.82	0.725	0.714	0.719	0.006	VDD/VSS
302.24	466.22	0.725	0.714	0.719	0.005	VDD/VSS
460.41	358.45	0.725	0.714	0.719	0.006	VDD/VSS
442.82	398.42	0.725	0.714	0.720	0.006	VDD/VSS
360.13	453.71	0.725	0.714	0.720	0.006	VDD/VSS
424.64	349.14	0.725	0.714	0.719	0.005	VDD/VSS
586.58	173.09	0.725	0.714	0.720	0.006	VDD/VSS
453.12	641.83	0.725	0.714	0.719	0.005	VDD/VSS
592.59	217.08	0.725	0.714	0.720	0.006	VDD/VSS

B. Signal Integrity Resolution

By simulating a design and verifying that the current flow for both power and signal lines do not exceed the foundry specified limits, we ensure that the interconnects and hence the design itself will meet required MTTF limits. EM violations on power nets and signal nets were successfully resolved by applying NDR and parallel routes while maintaining timing and area requirements.

C. Power Consumption Reduction

Mix-Vt optimization involves the strategic use of transistors with varying threshold voltages (Vt) within a circuit to balance power efficiency and performance. The threshold voltage of a transistor directly influences both its switching speed and power consumption—low-Vt transistors switch faster but consume more leakage power, while high-Vt transistors are slower but more power-efficient. By carefully assigning low-Vt transistors to critical timing paths that require higher speed, and high-Vt transistors to non-critical paths where power savings are prioritized, designers can effectively manage the trade-offs between power consumption and performance. This technique is especially valuable in advanced VLSI designs were achieving optimal PPA (Power, Performance, and Area) is crucial.

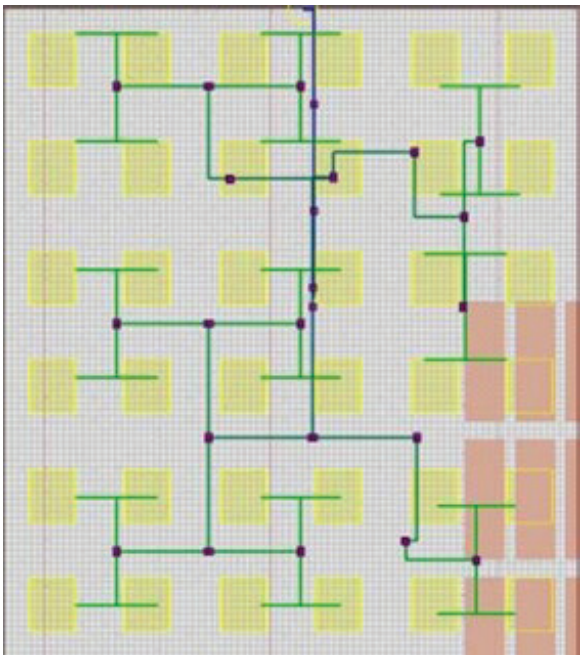


Fig. 1. Flexible H-Tree with D8 and D16 buffers

TABLE IV. POWER AND FREQUENCY DOMAIN ANALYSIS

Groups	Clock Pin Power (W)		Internal Power (W)	Leakage Power (W)	Switching Power (W)
	Power Domains:				
VDD	0.54692		0.72357	0.084563	0.54860
Total	0.54692		0.72357	0.084563	0.54860
	Frequency Domains:				
1.63e+09	0.51741		0.69259	0.081017	0.54705
1.56e+08	0.02944		0.029903	0.0012126	0.0015388
0	4.2677e-05		4.9056e-05	0.0023156	0
2.0e+08	1.9174e-05		2.7998e-05	4.7601e-06	2.1134e-05
1.563e+09	0		2.0617e-06	2.1027e-06	7.0672e-06
Total	0.54692		0.72258	0.084574	0.54862
	User Defined Groups:				
Comb. logic	0		0.12114	0.027283	0.37515
Seq. logic	0.33595		0.36186	0.0053713	0.044618
Memory	0.21096		0.21096	0.050704	0.0029605
Clock	0		0.028617	0.0007495	0.12589
Level shifter	0		0	0.00045495	0
Total	0.54692		0.72258	0.084574	0.54862

TABLE V.

Groups	Total Power (W)	Percent Power (%)
Power Domains:		
VDD	1.3558	100.00
Total	1.3558	100.00
Frequency Domains:		
1.563e+09	1.3207	97.41
1e+08	0.032654	2.41
0	0.0023647	0.17
2.5e+08	5.2652e-05	0.00
1.563e+09	3.3156e-05	0.00
Total	1.3558	100.00
User Defined Groups:		
Comb. logic	0.52358	38.62
Seq. logic	0.41185	30.38
Memory	0.26463	19.52
Clock	0.15525	11.45
Level shifter	0.00046497	0.03
Total	1.3558	100.00

Low- Vt transistors are used in critical paths or areas where high performance is required, while high-Vt transistors are used in non-critical or low-power areas. This approach allows for power savings by reducing leakage current and dynamic power dissipation in low-power regions, while maintaining the desired performance in critical regions.

Power consumption was optimized through the application of mix-VT optimization during standard cell placement and the implementation of a flexible H-Tree clock

distribution network. The H-Tree topology uses D8 buffers for leaf nodes and D16 buffers for trunks, balancing power consumption and performance. The overall area of the subsystem is 517,197.4432 um2 with 3032094 instances with a power consumption of 1.25 W.

TABLE VI. INSTANCE COUNT AND TOTAL AREA

Inst Count	Total Area
3,032,094	517,197.4432

VI. CONCLUSION

This study emphasizes significant progress in resolving physical design challenges related to memory controller subsystems in high-performance computing environments. The achieved reductions in IR drop, resolution of signal integrity issues, and power optimization techniques contribute to the development of more efficient and reliable high-speed VLSI circuits for sub-6nm nodes. The methodologies and techniques developed in this work have potential applications across a broad spectrum of high-performance, low-power VLSI designs for advanced computing systems.

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